

FSSAI: Food Safety and Standards Authority of India (FSSAI) is an autonomous body established under the ministry of health and family welfare, Government of India. FSSAI maintains the food quality levels in order to ensure safety and provides satisfaction to every consumer.

5.1.1 Classification of Foods

Foods may be classified as

a. Fresh foods

These are foods which have not been preserved and not spoiled yet. For example; vegetables, fruits and meat spoil immediately after harvesting or slaughtering.

b. Preserved foods

Foods are preserved by adding salt, sugar, acetic acids and ascorbic acids. Example: Jam, Pickles. In this way their shelf life is improved.

c. Canned foods

In canning, food products are processed and sealed in the air tight containers. It provides longer shelf life ranging from one to five years. Example: Baked beans, Olives.

d. Processed foods

During food processing, original nature of food is changed or altered. It is done by Freezing, Canning, Baking and Drying. Example: Breakfast cereals, Cakes, Biscuits and Bread.

e. Fermented food products

These foods are subjected to fermentation by the action of microorganisms. Example: Kefir, Cheese.

5.1.2 Sources of Microorganism in Food

The primary sources of microorganisms in food include,

1. Soil and water
2. Plant and plants products
3. Food utensils
4. Intestinal tract of human and animals
5. Food handlers
6. Animal hides and skins
7. Air and dust

5.1.3 Factors that Influence Growth of Microorganisms in Food

Many factors influence the growth of the microorganisms in food. Some of the factors are intrinsic and some others are extrinsic.

1. Intrinsic factors

The intrinsic factors include pH, moisture content, oxidation – reduction potential, nutrient status, antimicrobial constituents and biological structures.

a. pH: Every microorganisms has a minimal or maximal, and an optimal pH for its growth. Microbial cells are significantly affected by the pH of food because they apparently have no mechanism for adjusting their internal pH. In general, yeasts and molds are more acid tolerant than bacteria. Foods with low pH values (below 4.5) are usually not readily spoiled by bacteria and are more susceptible to spoilage by yeast and molds. Most of the microorganisms grow best at pH value around 7.0.

b. Moisture content: The preservation of food by drying is a direct consequence of removal of moisture, without which

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